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Comparison of Simple Auditory, Visual, and Audiovisual Reaction Times between Action Video Game Players and Non-Players

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Research Topic: Comparison of Simple Auditory, Visual, and Audiovisual Reaction Times between Action Video Game Players and Non-Players

Introduction and Research Problem

Reaction Time is the time taken between receiving a sensory stimulus and its respective behavioral response [1]. And auditory reaction time is usually found to be faster than the visual reaction time [1] [2]. But in E-sports, an electric sports which is rapidly growing worldwide, fast reactions are required at all times [3]. It has been established that action video game play enhances the domain of voluntary focus and spatial cognition, with promising signs for perception [4], as well as improved accuracy [5]. But no significant benefit was found from action video game play on auditory tasks [3] [5]. According to these findings, as visual processing is enhanced while auditory processing remains unchanged in action video game players (AVGPs), it would imply that the difference between auditory and visual reaction time should be smaller for these players. Although Stewart et al. [5] measured both visual and auditory reaction times [5], individuals were not tested for simple detection reaction times. Ersin et al. [3] measured simple reaction time on both visual and auditory reaction times [3], however, using different instruments (i.e. mouse press on visual and space bar press on audio) as well as different websites; moreover, groups were split at 14 hours a week play time mark rather than keeping the play time continuous, the sample was entirely male and contained no non-playing comparison group. Furthermore, on both studies, there was a lack of measurement on any combined audiovisual condition. This study aims to measure and compare reaction time to auditory, visual and combined audiovisual cues of AVGPs and non-players in a simple detection task. Both groups will be assessed in a single session using the same instrument to allow the difference between auditory and visual reaction times to be calculated within each participant, which are then compared between the two groups. Play time will be recorded continuously and related to the size of the auditory-visual difference for AVGPs. The results will suggest whether AVGPs should prioritize sound cues over visual cues and whether the difference in AVGPs will differ from non-players.

Literature Review

Simple detection reaction time test requires a person to responds as quickly as possible to a single stimulus [6]. Results found by Shelton and Kumar [1] shows that the visual reaction time is around 331 milliseconds and auditory reaction time is 284 milliseconds, which are mean values collected from a total of 14 participants [1]. Findings from Jain et al. [2] also shows similar results from 120 healthy medical students aged of 18 to 20 years, auditory reaction time was significantly less than visual reaction time; furthermore, they reported sex differences, with male medical students having faster reaction time for both auditory and visual stimuli than female medical students [2].

Study from Bediou et al. [4] presents a meta-analysis on perceptual, attentional, and cognitive measures of action video games from the year 2000 to 2015; it shows moderate advantage for habitual players over non-players in cross-sectional comparisons, the design used in current study ($g = 0.55$). Training intervention resulted in smaller advantage ($g = 0.34$), the advantage in cross-sectional comparison was similar for speed measures ($g = 0.61$) and accuracy ($g = 0.58$). This is interpreted by the authors as evidence against speed-accuracy trade-offs. They also found the published effects to be 30% larger than the full literature [4]. Stewart et al. [5], consisting of 80 participants aged 18 to 30 years, found that gamers have higher accuracy on a multiple object tracking task than non-players, but no differences in visual reaction time [5]. Findings by Ersin et

al. [3] shows that the 28 players above 14 hours a week out of a total of 53 participants, aged 19 to 27, were 51.54 milliseconds faster on visual reaction time and 138.22 milliseconds faster on the aim reaction time, which is a composite that includes movement time, ($p < 0.001$) than people playing less than 14 hours; tested on humanbenchmark.com [3]. From these studies, it has been established that there is a visual advantage in both speed and accuracy in pooled literature. However, the two studies closest to the design used in current study measured advantages using complex tracking tasks and compared players who differed in play volume; neither simple detection reaction test nor a gamer versus non-gamer comparison has been used.

In study by Stewart et al. [5], a Test of Attention in Listening (TAiL) presenting two successive pure tones and asking whether there has been change in frequency or location or remained constant, found no gamer advantage; moreover, the TAiL requires judgement from participants rather than responding to a single stimulus, which resulted in difference in scores indexing distraction and conflict resolution and not response to a single stimulus. Speech-in-noise tasks (i.e. BKB-SiN, LiSN-S) also found no gamer advantage, and falls under the same reason for requiring accuracy and not response speed. Consequently, the findings cannot be compared with simple visual reaction time to give a within-person modality difference [5]. Ersin et al. [3] measured simple auditory detection on playback.fm and found that groups did not differ in auditory reaction time ($p = 0.397$). However, comparison was between players of higher and lower volume of play time, both groups being active esports players, and not players versus non-players. Furthermore, two modalities were measured on different platforms, with their own timing characteristics, so values cannot be subtracted to give a within-person auditory versus visual difference [3]. Hamzeloo et al. [7] tasked 30 AVGPs and 30 non-players in experiment 1to search video clips of realistic scene for a target object while hearing a sound that was congruent with the target, congruent with a distractor, or neutral. The results showed 220 milliseconds advantage, 14% difference for AVGPs, but no difference between 36 AVGPs and 34 non-players in experiment 2; they stated that the null results were inconsistent between two experiments. The main effect of group was not significant and they concluded a reasonable doubt on whether action video games transfers to search in naturalistic multisensory scenes. However, this task tests whether meaning-based cross-modal cues speed up search and not a response to a sound, modality cannot be compared as there was no auditory-only condition [7]. Across these studies, two modalities have not been measured within the same participant on a common instrument.

Redundant Signal Effect (RSE), responses to combined auditory and visual signals, are faster than to either component alone; this effect has been studied across modality paring and replicates reliably with varying sample sizes across studies [8]. Donohue et al. compared video game players and non-players from 45 male participants on simultaneity and temporal-order judgement tasks, in both tasks video game players were found to be able to determine audiovisual timing more precisely [9]. Hamzeloo et al. found gamers used semantically congruent cross-modal cues as effectively as non-players, but in a search task rather than speeded detection [7]. Across these studies, whether the audiovisual redundancy in simple audio and visual detection differs between players and non-players has not been tested.

Research Question and Hypothesis

Research Question 1: Does the difference between auditory and visual simple reaction time differ between action video game players and non-players?

Hypothesis 1: The difference will be smaller in players as action gaming is associated with enhanced visual performance including reaction time [3] [4], while auditory performance shows no comparable advantage [3] [5] [7], so the auditory lead should narrow in this group.

Research Question 2: Do combined audiovisual cues produce faster responses than the faster of two single cues, and does this redundancy gain differ between groups?

Hypothesis 2: A redundancy gain will be present in both groups [8], but will not differ between them, since video game players have shown null for more efficient use of cross-modal information [7].

Research Question 3: Within players, does play time predict the size of the difference between auditory and visual simple reaction times?

Hypothesis 3: Greater play time will predict a smaller difference following the same reasoning as hypothesis 1.

If the difference between auditory and visual simple reaction time in two groups are the same and shows no relation to play time, the statement that visual enhancement narrows the modality gap would not be supported.

Methodology

Design: This study will use a quantitative quasi-experimental mixed design. Cue type (i.e. auditory, visual, combined) within participants, grouped as AVGPs and non-players between participants. Median reaction time per condition and error rates will be measured. Weekly play time will be recorded continuously within the player group.

Participants: A total of 100 students aged 15 to 18 will be recruited from a high school in Myanmar. Participants will be grouped into 50 AVGPs who play action video games for 5 or more hours a week in the past year, and 50 non-players who play action video games for less than 5 hours a week across all game categories in the past year [5]. All volunteers will be asked to complete a short screening questionnaire before testing; groups will be assigned and weekly hours and years of play will be recorded. Recruitment will be stratified across four cells defined by group and sex to a target of 25 per cell [7], since sex has been associated with reaction time [2] and has been confounded with group in previous samples, in sample by Stewart et al., the player group comprised 13 males out of 15, and the non-player group 16 females out of 16 [5]. If cells cannot be filled, the composition will be reported and sex entered as a covariate. Participants will be required to report normal hearing and normal or corrected-to-normal vision [8], and remain naive to the study's aim until debriefing. The sample size of 50 per group exceeds the group sizes used in comparable studies (i.e. Stewart et al., Ersin et al., Hamzeloo et al.) and is the largest sample feasible for in-person testing by a single researcher.

Apparatus: All testing will be conducted on a single Dell laptop with a 13-inch screen at 60 Hz, in a quiet room at the school, with viewing distance held at approximately 57 cm [8]. Auditory stimuli will be presented through wired over-ear headphones at a fixed volume, identical for

every participant. The task will be carried out in PsychoPy, which is designed for millisecond accuracy of stimulus presentation and response collection. Responses will be made via a single press of space bar on the laptop keyboard.

Stimuli: The stimuli will be presented on a grey background with a small green point at the center of the screen, visible throughout each trial. The visual signal will appear as a white ring of 1 cm diameter centered on the fixation point, displayed until a response is made or for a maximum of 1.5 seconds. The auditory signal will be a 440 Hz pure tone on both ears at 45 dB SPL with a 10 milliseconds onset ramp, for the same maximum duration [8]. In combined condition both signals will be present with synchronized onsets and on catch trials no signal will be presented and participants must withhold their response. Detectability will be checked in a pilot before the main study begins.

Procedure: Participants will be tested individually in a quiet room and they will be instructed to respond as quickly as possible by pressing the space bar key on the laptop keyboard whenever they detect a signal, and to withhold their response when no signal appears. A practice block of 15 trials with feedback will be conducted beforehand and will not be analyzed. The main task will consist of 5 blocks of 52 trials; each block containing 13 auditory, 13 visual, 13 combined, and 13 catch trials in randomized order, giving 65 trials per signal condition. Each trial will begin with the fixation point, followed by a delay of 1 second plus a random interval averaging 0.75 seconds, after which the signal appears for up to 1.5 seconds or until a response [8]. Feedback will be shown following anticipations, misses and false alarms; participants may take a short break between blocks with the full session lasting approximately 20 minutes. Five sessions per day will be carried out over 5 school days a week, the procedure will therefore be carried out for approximately 4 weeks.

Measures: The median reaction time in milliseconds will be calculated for each participant in each of the three signal conditions, measured from signal onset to key press; reaction time distributions are positively skewed and medians are less affected by occasional slow responses than means. The difference between auditory and visual reaction time is then calculated, computed per participant as median visual minus median auditory reaction time, where a positive value indicates an auditory advantage. Only correct responses will contribute to the median; anticipations (i.e. responses faster than 150 milliseconds), misses (i.e. no response within 1.5 seconds), and false alarms on catch trials will be excluded from the data, these data will be counted and calculated respectively as a percentage of total number of trials. Participants with a false alarm rate above 15% on catch trials will be excluded, as this indicates the participant guessing rather than detecting. From screening questionnaire, weekly hours of action game play, years of play, genres played, and objective playtime from game platforms such as Steam or from console profiles where the participant can access it will be recorded for research question 3.

Data Analysis:

JASP software will be used. A 3-cue type by 2 groups mixed ANOVA, with the interaction as the primary test. For the focused comparison, the auditory-visual difference score is computed per participant and compared between groups with an independent-samples t-test, reported with Cohen's d and a 95% confidence interval.
The redundancy gain is computed per participant as “the faster unisensory median minus the combined median”, tested against zero to confirm a gain is present, then compared between groups.

Correlation between weekly play time and the auditory-visual difference score within the player group will be checked.

Bayes factors (BF_{10}) will be reported alongside each conventional test using default priors, BF_{10} below 1/3 will be taken as evidence for the null hypothesis for hypothesis 2 and BF_{10} above 3 as evidence for the alternative, with intermediate values treated as inconclusive [5] [7].
The hypothesis, exclusion criteria, and analysis plan will be preregistered on the Open Science Framework before data collection begins and any analyses not specified in advance will be reported as exploratory.

Ethical Considerations: The study will be conducted with the permission of the school principal and a member of staff supervising. Written informed consent will be obtained from all participants, with parental or guardian consent in addition for those under 18. Participation is voluntary, participants are allowed to withdraw at any point without giving a reason and they may request that their data be removed. Risks involved are minimal, auditory stimuli are presented at 45 dB SPL, well below levels associated with hearing risk, and breaks will be provided between blocks to limit fatigue. Participants will be identified by number only to keep anonymity, no names are stored with the data, files are kept securely, and results will be reported in aggregate. The study's aim will not be disclosed before testing to keep participants naive, and a full debriefing will be given afterwards.

Limitations: Quasi-experimental with no random assignment could result in any group difference being an association and could reflect self-selection into gaming rather than an effect of it. The sample of 15- to 18-year-olds from one school limits generalization to adults or other populations. Play time is self-reported and subject to bias [5], partly mitigated by collecting objective playtime records where participants can access them. A standard keyboard and a 60 Hz display introduce a small constant delay, this applies equally to all conditions and so it does not bias the within-participant comparisons, but absolute values should not be compared with those from other studies. Effects smaller than about $d = 0.57$ may not be detected, so non-significant results will be interpreted with Bayes factors rather than treated as evidence of no difference.

Expected Outcome

Research Question 1: If the auditory-visual difference is smaller in players, that supports modality-specific enhancement, where visual processing speeds up and auditory does not; the extreme case of smaller is the reversed advantage of visual over auditory reaction time. If the difference is equivalent in both groups, the visual advantage reported by Bediou et al., Stewart et al., and Ersin et al. does not extend to simple detection, which sets a boundary on where gaming-related advantages appear.
Research Question 2: Redundancy gain is expected in both groups. If it doesn't differ between them, that extends Hamzeloo et al.'s null from semantic cross-modal cueing to speeded detection.
Research Question 3: A gradient with play time would support the training account. On the other hand, its absence, given the restricted range, would be uninformative rather than contrary evidence.

Contribution: This would be the first within-person comparison of auditory and visual detection in gamers on a common instrument, and the results bear on whether players should prioritize auditory cues and whether game designers should pair important events with both a sound and a visual signal.

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